TESS-W Firmware Upgrade

1. Introduction

The TESS-W firmware upgrade is an exceptional procedure that must be made whenever a serious firmware bug is discovered and it is not practical to return the device to the manufacturer for service.

The outline of the process is

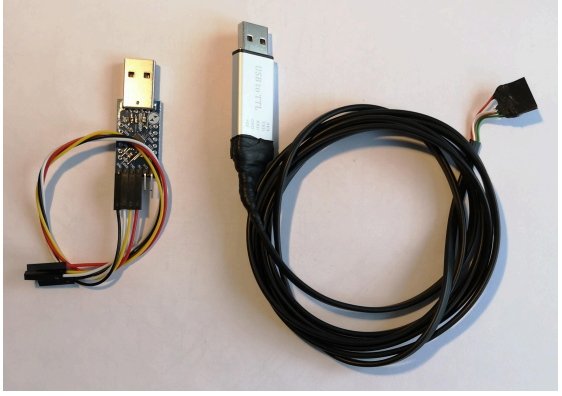
- Open the device.
- Perform the firmware upgrade.
- Close the device.

To update the firmware you will need:

- A Philips Screwdriver.
- A micro-USB cable or custom UST-to-TTL cable depending on the printed circuit board (PCB) version.
- The firmware file or files. Normally you will need only one file for a simple upgrade process retaining the configuration.
- The uploader program.
- A PC running Windows or Linux.

The custom USB-TTL cable is connected to the J8 connector in the printed circuit board (PCB). Pinout in table below.

PCB TESS-W **J8 PROG connector**		**Adapter** **USB TTL**
1	5v	5v
2	not connected	See image below
3	TX	RX
4	RX	TX
5	GND	GND



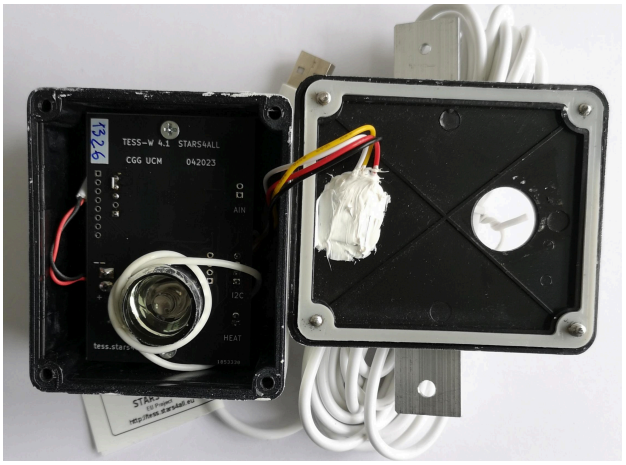


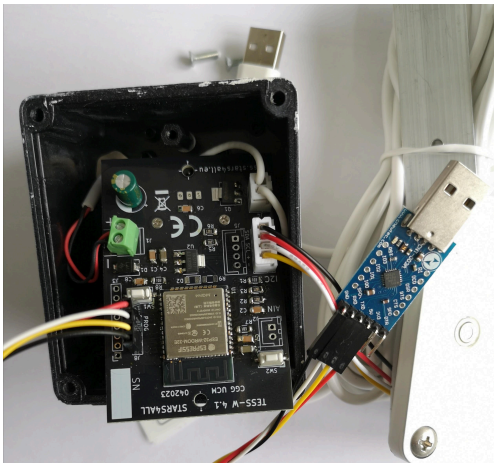
Some Custom USB-TTL cables (left). Detailed image (right) shows pin 2 not being used.

2. Open and connect the adapter.

It is necessary to open the photometer to connect the USB cable.

- Remove the cover by unscrewing the top four screws.
- Remove the two screws from the PCB to access the components
- Connect the USB-TTL adapter. It is not necessary to connect the USB power cable. The device is powered by this adapter, no need to connect the ordinary USB power cable.
- Check with a serial terminal that the photometer is connected and giving readings (section 4.). If no readings are received, the USB-TTL adapter driver must be installed manually (section 6.).

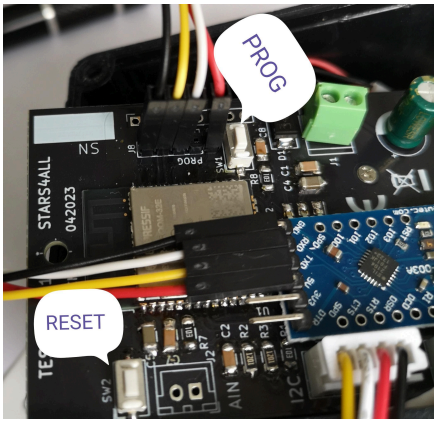




3. Firmware update.

3.1. Set device in Programming mode

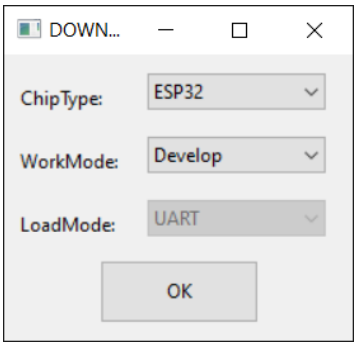
If using the custom USB-TTL cable the firmware upload requires first pressing the PROG button and before releasing it, briefly press the RESET button. On a board with a micro-USB connector, this procedure is not necessary.



The module will enter into programming mode.

3.2. Firmware Update (Windows)

Download and unzip the update package from:
https://observatorioremoto.com/TessW_downloadtool.zip
It contains the **update tool**, the **binary files** to be loaded and a small terminal program called **Termite** for testing.

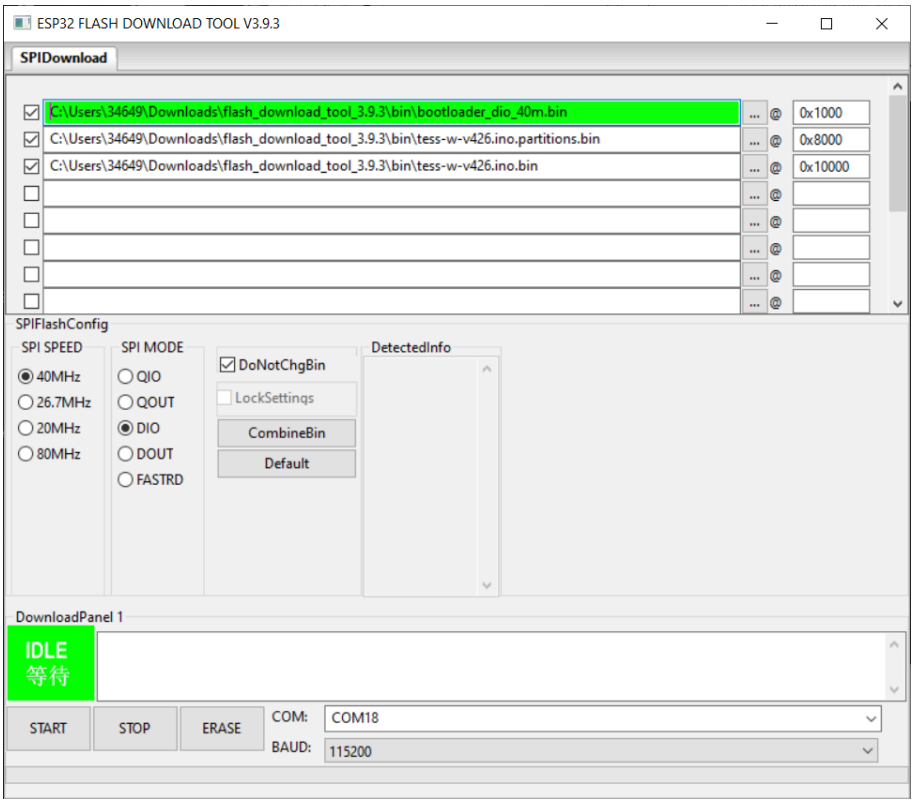


Before starting the update, use Termite and check how the photometer works. This test is described in section 4.
Do not forget to close Termite to free the serial port.

Launch the upload program by clicking on the executable ***flash_download_tool_3.9.3.exe***

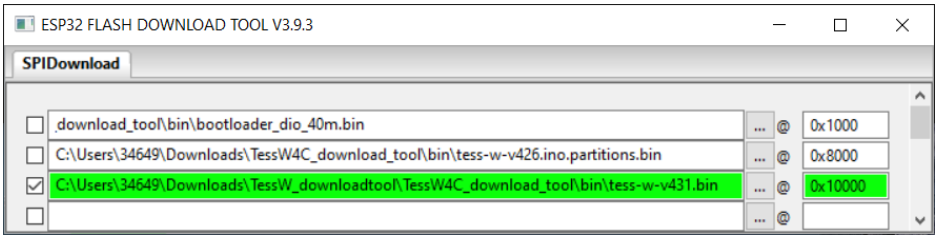
Select Chip Type: ESP32 and press OK.

The main window appears showing three binary files located in the \bin folder.

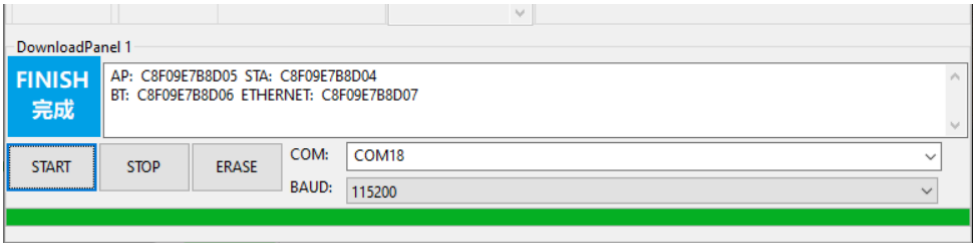


The path to the files can be chosen from the button on the right of the file. They should be changed if they are on another path.

Three files are shown, which are required for new installations. Only the third one is required for the update:



Select the COM port assigned to your TESS.
In the photometer, press the PROG button and before releasing it, briefly press the RST button.
Then press the START button in the program.
Final check status if all went well:



Close the Download Tool.
Relaunch the terminal Termite to verify the update.

3.2 Firmware update (Linux)

The following procedure describes how to update the firmware using the Linux Command Line Interface.

You will need a modern Python 3 distribution (>=3.7), the pip and virtual environment creation Python utilities. You will also need a terminal emulator using a serial port. Consult the help pages relative to your particular distribution. The example assumes an Ubuntu environment with `python3-pip` and `python3-venv` packages already installed.

- Prepare a Python virtual environment and activate it

```
$ mkdir esptool
$ cd esptool
esptool$ python3 -m venv .venv
esptool$ source .venv/bin/activate
```

- Install the ESP32 flash tool

```
(.venv) esptool$ pip install esptool
```

Also make sure that the firmware file (i.e. **`tess-w-v431.bin`**) is in the same directory.

- Prepare the device to be programmed
  Make sure that you have set the device as in section 3.1 Set device in programming mode. This step is skipped when using and ordinary micro-USB cable.

- Connect the cable to the device and host computer.
  If using a micro-USB cable, you should see a new ACM device:

```
(.venv) esptool$ ls -la /dev/ttyACM*
crw-rw---- 1 root dialout 166, 0 oct 11 15:08 /dev/ttyACM0
```

If using a custom USB-TTL cable, you should see <NOT YET COMPLETED>

- Update the device using esptool.py

```
(.venv) esptool$ esptool.py --chip auto --port
/dev/ttyACM0 write_flash 0x10000 tess-w-v431.bin
--flash_freq 40m --flash_mode dio --verify
```

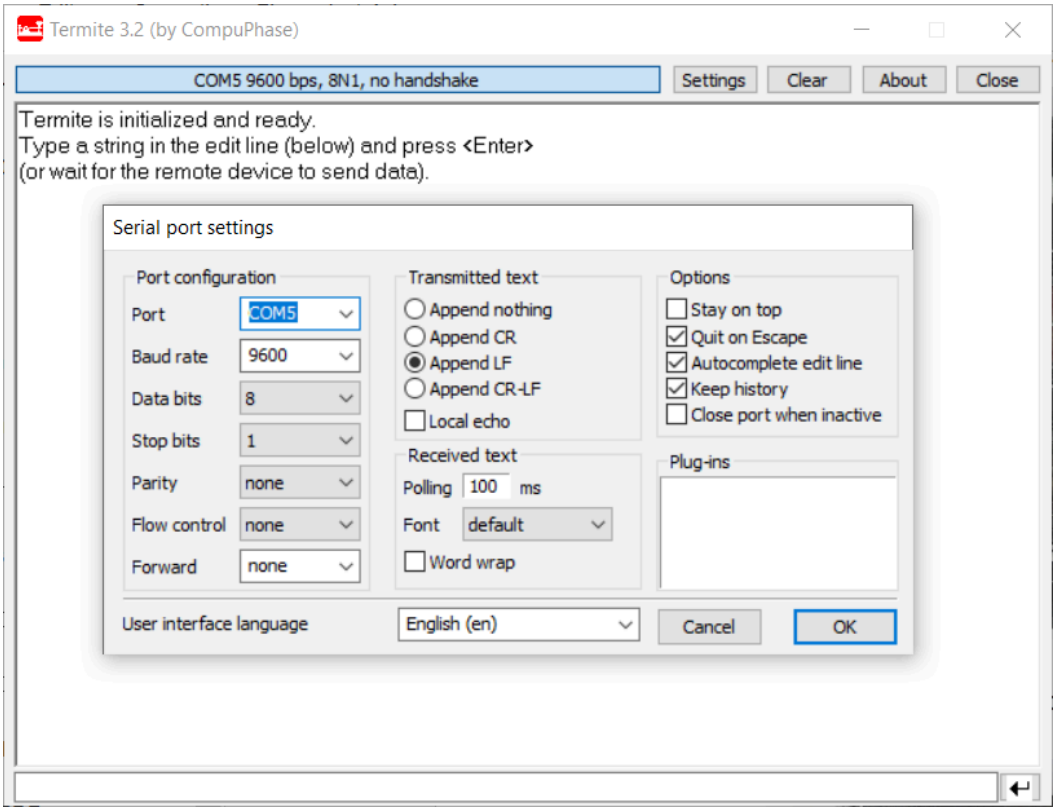
(this is a single long line)

4. Testing via the USB cable.

4.1. Testing on Windows

Using the USB cable connected internally to the photometer, and with a serial terminal, we can observe its operation.
For windows I recommend Termite (included), which consists of a single file.

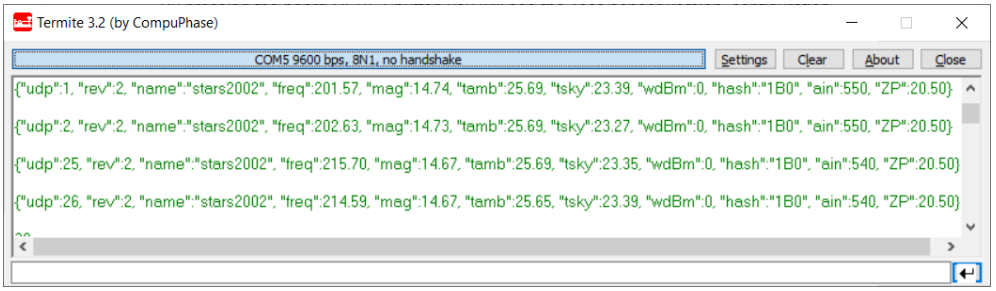
After clicking Termite.exe, go to Settings and select the COM port assigned to your TESS.



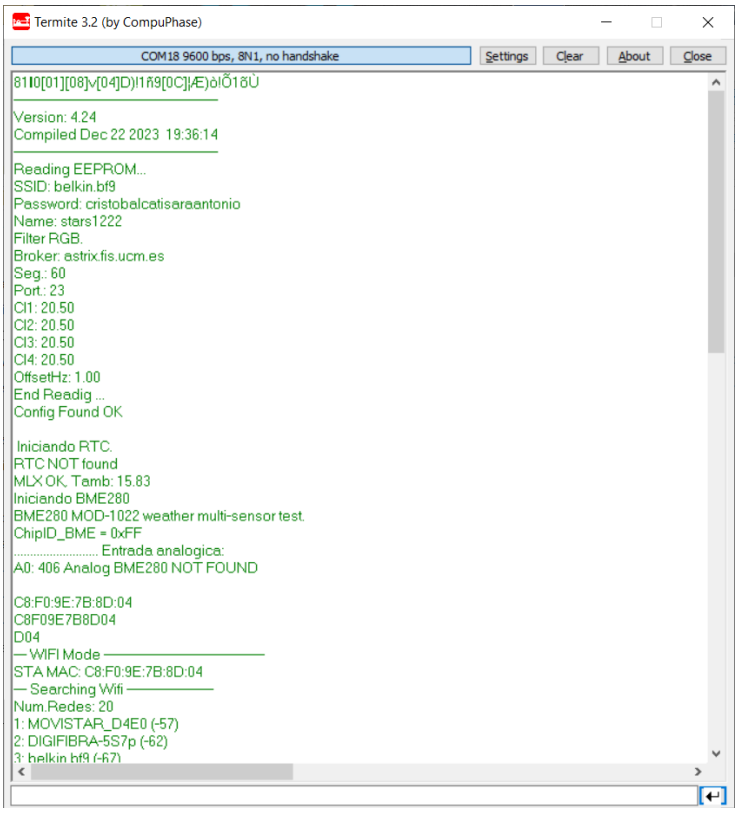
Once the terminal is connected we will see its messages.

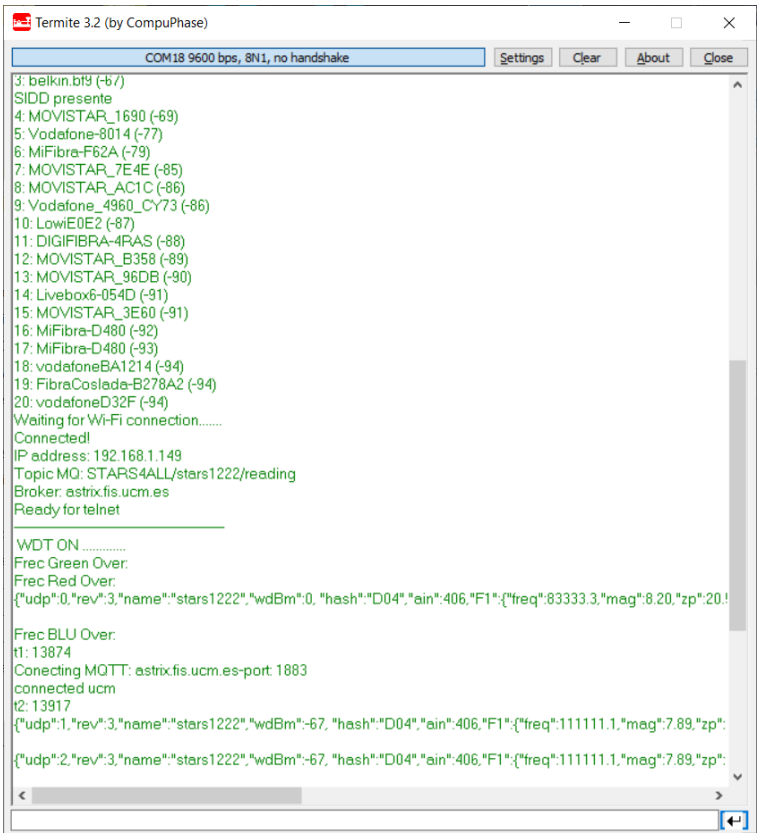
In order to read the information, we can stop and start the data flow by clicking on the top bar, which indicates the COM port.

In aTESS-W with configuration prior to the update, we can see something like this:

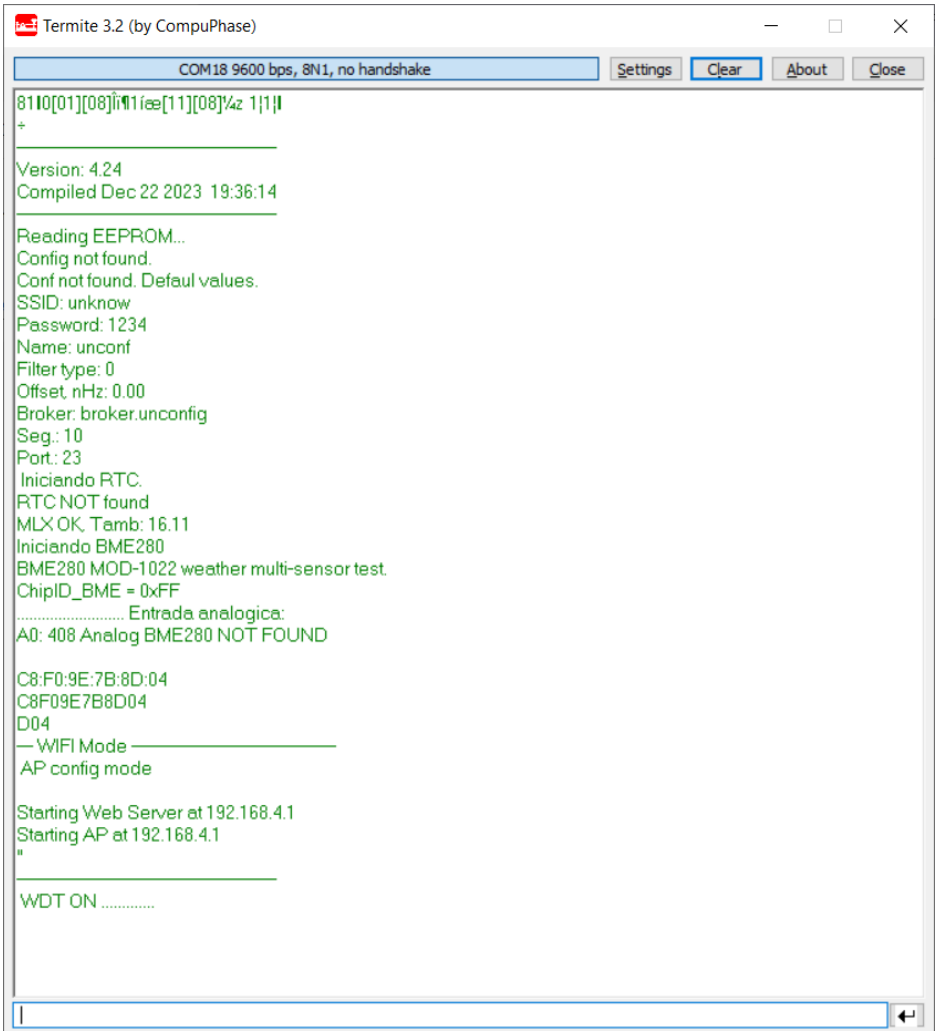


By pressing the Tess RESET button you will see the Tess sensor versión, configuration, the beginning of the program and finally the readings.





If the TESS-W had no previous configuration or had been deleted with a Full Reset we will see this:

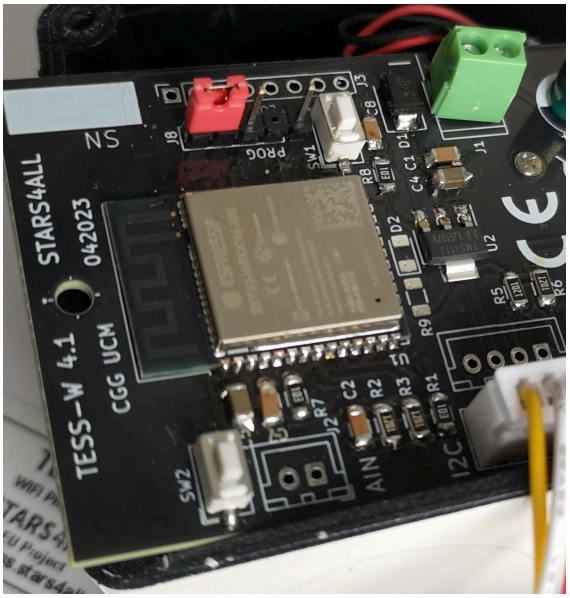


4.2. Testing on Linux, command line

You can use whichever tool (minicom, screen, etc) to connect to the device via a serial line port. In the example we use the **`screen`** utility.
The serial line communication parameters are: 8 bits, 9600, no parity, no hardware control flow

```
screen /dev/ttyACM 9600
```

4.3 Remove serial cable & plug jumper.



On PCB version 4.1 it is recommended that the Rx pin of the programming input be jumpered to ground. Red jumper in the image.

5. Closing the device.



You should close the device using the same Philips type screw-driver. It must be firmly closed to seal the device and protect it from the environment. All four screws should be closed applying the same pressure. Apply just enough force to each screw so that the cover is slightly curved on all four sides as shown in the photo.

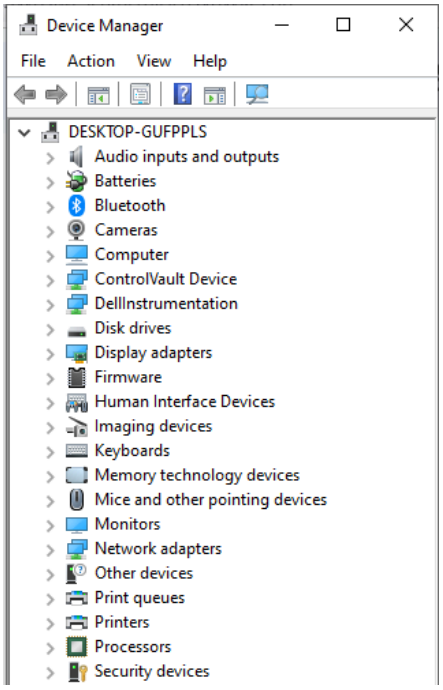
As seen in the picture, the aluminum strip serves as a ruler and shows the cover slightly bent.

6. Windows USB driver installation.

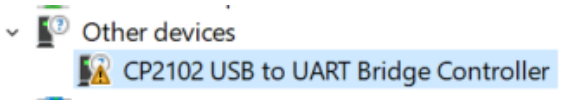
It may happen that the first time we connect the photometer by USB, Windows does not find the driver for the USB to serial adapter. Then we have to search for it and download it manually.

To download the driver we first need to know which chip is in our photometer.

In Windows, with the photometer connected to our PC, open the **Device Manager:**



and cllick on **Other devices**, it will show the chip installed in Tess. It is for that chip that we need to look for the driver on internet:



The most common USB-TTL adapter is from SILABS. On their website you can download “CP210x_Windows_Driver”.
Once unzipped, we will execute the installer. The installation must be done as an administrator user.

